

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter 2022

Date: 12/05/2023

Course : B. Pharmacy
Subject Name : Pathophysiology
Max Marks : 75

Sem : II
Subject Code : BP204T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all questions) (10 x 2) = 20

- i) Define terms leprosy and depression
- ii) Describe signs and symptoms of Parkinson's disease.
- iii) Define iron deficiency anemia and megaloblastic anemia.
- iv) Enlist signs and symptoms of hypertension.
- v) Enumerate causes of acute renal failure.
- vi) Enlist any four disorders related to sex hormones.
- vii) Enlist signs and symptoms of rheumatoid arthritis.
- viii) Differentiate between hyperthyroidism and hypothyroidism
- ix) Enumerate signs and symptoms of urinary tract infection.
- x) Discuss the etiology of jaundice

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What are metabolic disorders? Explain the pathophysiology of diabetes mellitus.
- ii) Explain the etiology & pathogenesis of peptic ulcer.
- iii) Explain the etiology, pathogenesis and morphology of reversible cell injury.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain pathophysiology of gout.
- ii) Enlist and explain types of hepatitis.
- iii) Discuss pathophysiology of epilepsy.
- iv) Explain the pathogenesis and clinical symptoms of Asthma.
- v) Discuss chemical mediators of inflammation.
- vi) Explain etiology and pathogenesis of tuberculosis.
- vii) Explain the pathophysiology of AIDS.
- viii) Explain the pathogenesis and clinical symptoms of Chronic Obstructive Airway Disease.
- ix) Discuss in brief about atherosclerosis.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 03/05/2023

Course : B. Pharmacy
Subject Name : Human Anatomy and Physiology-II
Max Marks : 75

Sem:
Subject Code :
Duration :

II
BP201T
3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

(10 x 2) = 20

Q. 1. Objective Type Questions (Answer all the questions)

- i) Define the terms Synapse and Neuroglia.
- ii) Define Neurology and name the parts of CNS.
- iii) Define the terms Peristalsis and Achlorhydria.
- iv) Draw neat labeled diagram of Stomach.
- v) Define the terms Tidal Volume and Vital Capacity.
- vi) Define the terms glomerular filtration rate and Filtration fraction.
- vii) Define the terms hyperplasia and ectopic pregnancy.
- viii) Define the terms gynaecomastia and hypoglycaemia
- ix) What is cushing's syndrome.
- x) Enlist different phases of Menstruation Cycle.

(2 x 10) = 20

Q. 2. Long Answers (Answer 2 out of 3)

- i) Describe in detail about glomerular filtration and tubular reabsorption process.
- ii) Describe the structure and functions of Liver and explain various disease associated with it.
- iii) Discuss in detail female reproductive system.

(7 x 5) = 35

Q. 3. Short Answers (Answer 7 out of 9)

- i) Define action potential of neuron and explain in detail generation of nerve impulse.
- ii) Write a short note on hypothalamus.
- iii) Describe Functional Organization of Cerebral Cortex.
- iv) Draw the structure of testes and explain the spermatogenesis process.
- v) Discuss the mechanism of external respiration and factor affecting gaseous exchange.
- vi) Explain the process of absorption of food from small intestine.
- vii) Write a short note on Pituitary gland.
- viii) Describe the anatomy of thyroid gland and enlist name and functions of their hormones.
- ix) Discuss in detail the structure of neuron.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter 2022

Date: 09/05/2023

Course: B. Pharmacy

Subject Name: Pharmaceutical Organic Chemistry-I

Max Marks: 75

Sem: II

Subject Code: BP202T

Duration: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Give structure and uses of Ethanolamine and Benzoic acid.
- ii) Enlist any two qualitative tests for carboxylic acids.
- iii) What is saytzeff's rule? Give its reaction.
- iv) Write in short stability of alkene.
- v) Define functional isomerism with example.
- vi) Give two examples for nucleophilic addition reaction to carbonyl compounds.
- vii) Draw the structure and write uses of Iodoform.
- viii) Draw structure and write uses of Propylene glycol.
- ix) Draw the structure and give IUPAC name of isobutanol.
- x) Enlist any four reactions of alkyl halide.

Q. 2. Long Answers (Answer 2 out of 3)

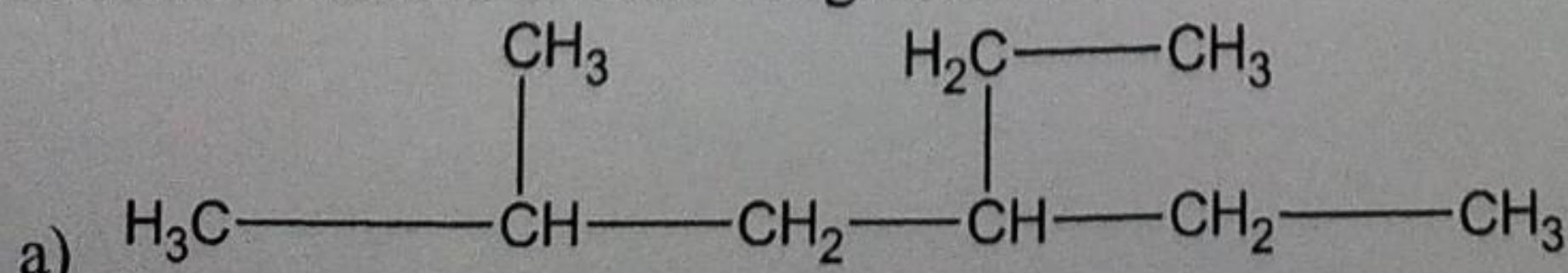
(2 x 10) = 20

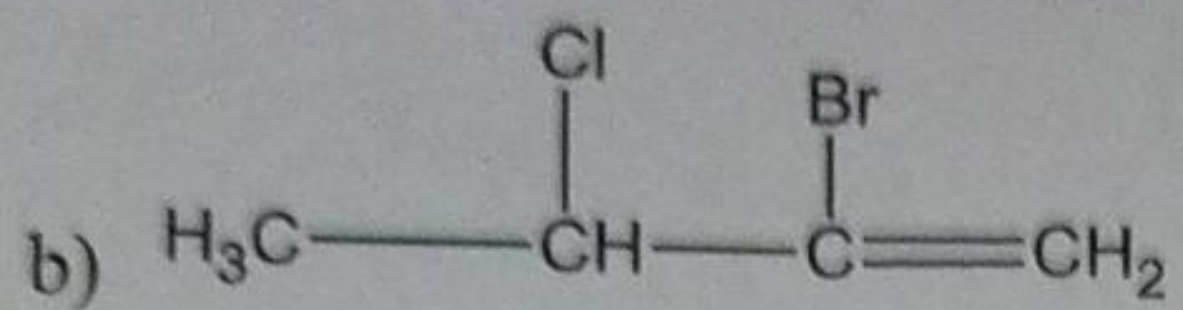
- i) Illustrate the mechanism of Markownikoff's and anti- Markownikoff's orientation of alkene with suitable examples.
- ii) Explain the reaction and mechanism of Perkin and Aldol condensation.
- iii) Explain sp^2 hybridization in alkenes. Write a note on Ozonolysis and Diels-alder reaction.

Q. 3. Short Answers (Answer 7 out of 9)

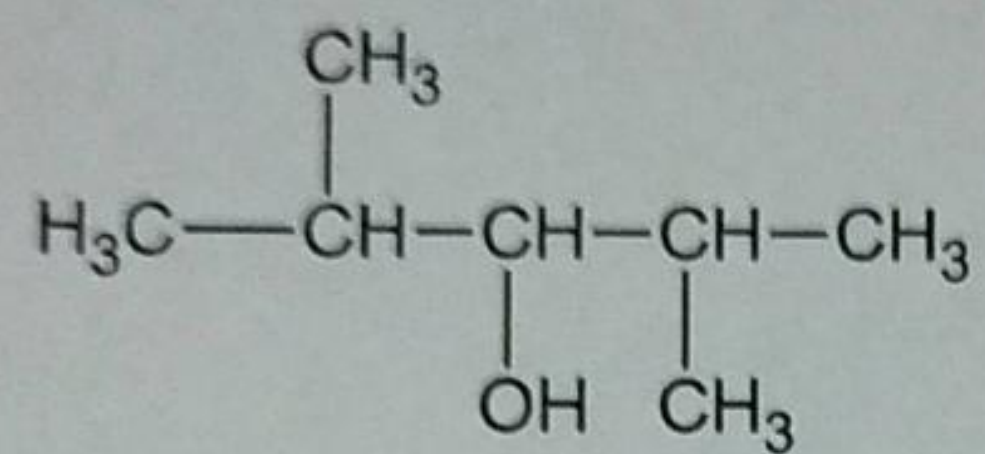
(7 x 5) = 35

- i) Explain the sp^3 hybridization in alkanes with suitable example.
- ii) Classify organic compounds with example.
- iii) What is effect of substitution on acidity of carboxylic acid?
- iv) Explain the factors affecting $E1$ reaction.
- v) Give an account on Cannizzaro and cross-cannizzaro reaction.
- vi) How you will identify alcohol and aliphatic amines by qualitative tests?
- vii) Discuss the free radical addition reactions of conjugated dienes.
- viii) How you will synthesize alkenes?
- ix) Write IUPAC names of following structures

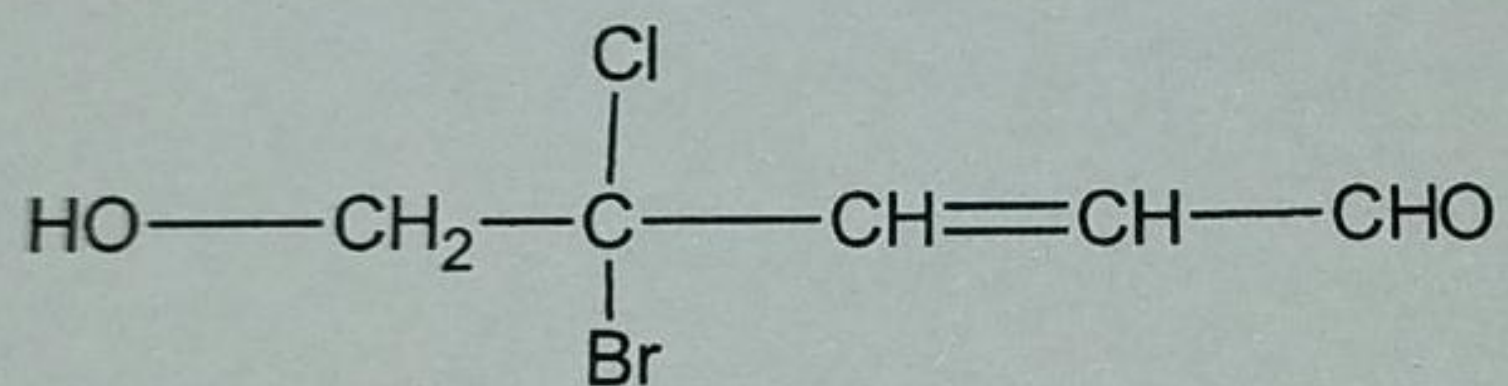




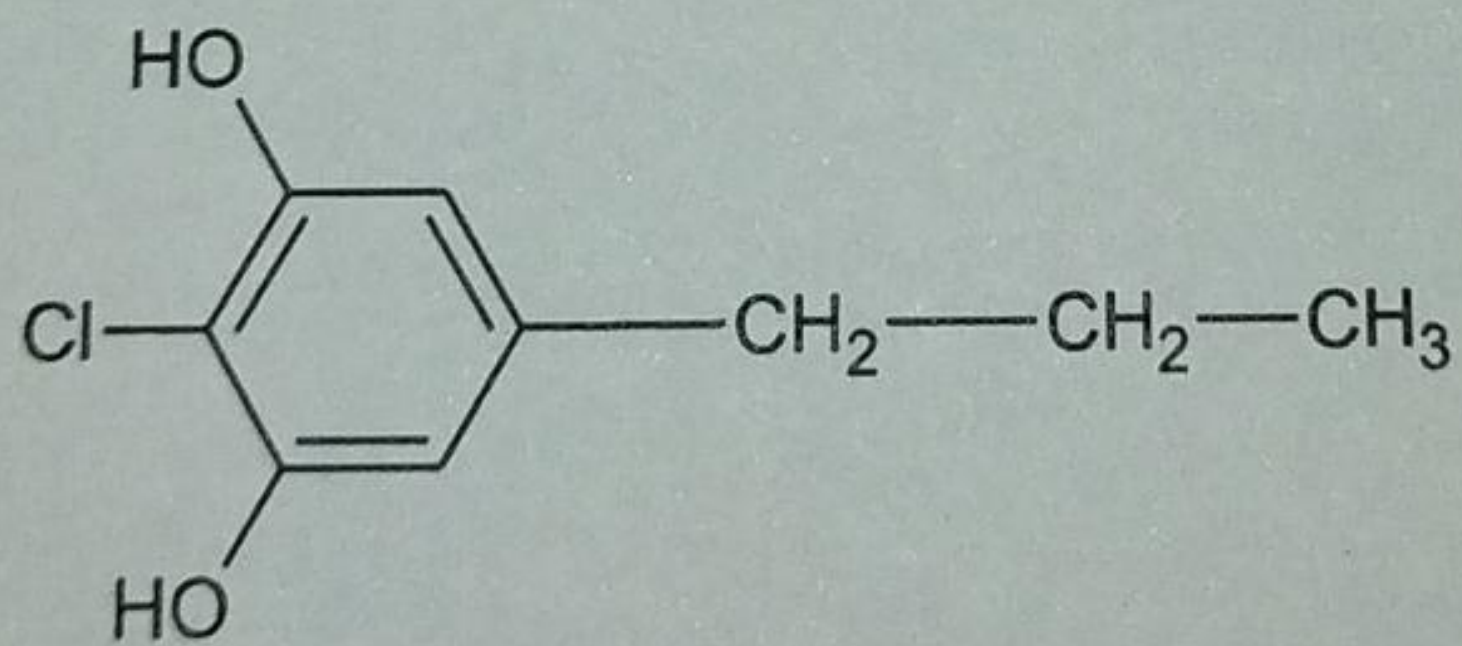
c)



d)



e)



-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 11/05/2023

Course	: B. Pharmacy	Sem	: II
Subject Name	: Biochemistry	Subject Code	: BP203T
Max Marks	: 75 Marks	Duration	: 03 Hrs.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 × 2 = 20)

- i) What is OTHLIL? Explain in Short.
- ii) Enlist examples of inhibitors of protein biosynthesis?
- iii) Explain in short about "Ketoacidosis"
- iv) Write full forms of any four
 - a) ATP
 - b) NADH
 - c) NADPH
 - d) FMNH₂
 - e) FAD
- v) Write any four functions of proteins with examples.
- vi) Define & explain in short about Coenzymes.
- vii) Draw Correct double helix structure of DNA
- viii) Mention different metabolic disorders associated with Urea Cycle?
- ix) Write names of any **Eight** enzymes involved in Glycolysis
- x) Explain in short about any two polysaccharides

Q. 2. Long Answers (Answer 2 out of 3)

(2 × 10 = 20)

- i) Classify carbohydrates with suitable examples, what are different steps involved in TCA cycle? Explain with its energetics.
- ii) Explain in detail about β-oxidation of fatty acids (Palmitic Acid).
- iii) What are biomolecules? Write definition, classification & biological functions of biomolecules.

Q. 3. Short Answers (Answer 7 out of 9)

(7 × 5 = 35)

- i) Write a note on "Energy Rich Compounds".
- ii) What are Lipids? Write Classification with examples and biological role of Lipids.
- iii) Write in brief about "Glycogenolysis".
- iv) Explain about inhibitors of ETC.
- v) Write a short note on "Disorders associated with Tyrosine Metabolism".
- vi) Explain Stages involved in RNA Transcription.
- vii) Write a short note on Hyperuricemia & Gout disease.
- viii) Explain Enzyme Kinetics with focus on "Effect of concentration of substrate".
- ix) Write a short note on Diabetes Mellitus.

-----END OF THE PAPER-----